

ASSIGNMENT 3:

Module 4: Functions & Modules in Python

Task 1: Calculate Factorial Using a Function

Problem Statement: Write a Python program that:

1. Defines a function named factorial that takes a number as an argument and calculates its factorial using a loop or recursion.
2. Returns the calculated factorial.
3. Calls the function with a sample number and prints the output.

Expected Output:

For example, if the function is called with 5, it should return:

```
Enter a number: 5  
Factorial of 5 is: 120
```

Task 2: Using the Math Module for Calculations

Problem Statement: Write a Python program that:

1. Asks the user for a number as input.
2. Uses the math module to calculate the:

- Square root of the number
 - Natural logarithm (log base e) of the number
 - Sine of the number (in radians)
3. Displays the calculated results.

Expected Output:

For example, if the user enters 25, the output should be:

```
Enter a number: 25
Square root: 5.0
Logarithm: 3.2188758248682006
Sine: -0.13235175009777303
```

Submission Instructions:

- Keep all required files in **one single folder**.
- Make sure all files are complete and correctly named.
- Convert the folder into a **ZIP file** before uploading.
- Rename the ZIP file using your **Name_AssignmentName** format.
- Check the ZIP file once to ensure all files open properly.
- Upload **only the ZIP file** on the submission portal.
- Add both **Task 1 and Task 2** scripts in the same Folder.
- Submit the **Zip File**.

Reference:

Follow the **Python course - Module 4: Functions & Modules in Python** for additional guidelines and examples.

Note:

Please test your project thoroughly and check all the validations and error handling prior to ensure it works as expected before submission.

You can always connect to the mentor using the chat support option for any doubts or queries!